Code Change Summary - MedScan AI

BBIS 7502 Applied Project | Saroj Kumar Tamang | Student ID: 108579

Baseline Source

Item	Details
Repository	https://github.com/simranjeet97/Awsome_AI_Agents
Author	Simranjeet Singh (simranjeet97)
Licence	Apache 2.0
Branch	main
Folder reviewed	GenAI_Medical_Imaging_Agent
File reviewed	app.py
Framework	Agno
AI Model	Google Gemini 2.0 Flash
Interface	Streamlit
Reference source	DuckDuckGo
Output format	Plain unstructured text
Authentication	None
Database	None

Changes Made During Extension

Component	Change Made
Agent framework	Removed Agno and replaced with direct calls to Google GenAI SDK to resolve API compatibility er
AI model	Upgraded from Gemini 2.0 Flash to Gemini 2.5 Flash after free tier quota errors.
Prompt design	Redesigned from a short basic instruction to a detailed structured prompt requiring seven specific s
Authentication	Added Google OAuth 2.0 login via Streamlit st.login() so only authenticated users can access the s
Database	Added two table MySQL database schema storing user accounts and analysis history linked by a f
Report display	Added regular expression parser to split report into sections and display each as a colour coded ca
References	Replaced DuckDuckGo with PubMed Entrez API for general contextual biomedical references.
Urgent alert	Added experimental keyword based urgent findings alert displaying a red warning banner.
Chatbot	Added follow up chatbot in a separate tab grounded in the specific report context.
PDF export	Added PDF export feature using ReportLab allowing formatted report download.
Sidebar history	Added sidebar displaying each user's ten most recent analyses with modality and timestamp.
Image processing	Added Pillow image resizing before sending to Gemini API.
Safety	Added mandatory disclaimer section in every generated report.
Multi-user support	Added per user data separation via foreign key relationship in database.

Code Not Changed

Component	Reason Kept
Streamlit	Retained as the web interface framework from the baseline
Google Gemini	Retained as the underlying AI model, upgraded to newer version only
Image upload	Basic upload functionality retained and extended
Single file structure	Overall single application file structure retained